

13.4.2 A macroscopic perspective

Consider turbulent diffusion from a more macroscopic point of view, and in particular consider the transport of a nearly materially conserved tracer obeying

$$\frac{D\phi}{Dt} = D, \quad (13.66)$$

where the advecting flow is incompressible and D is a dissipative process such that $\overline{D\phi} \leq 0$ (a conventional harmonic diffusion has this property). By decomposing the fields into mean and eddy components in the usual way an equation for the evolution of the tracer variance can be straightforwardly derived, namely

$$\frac{1}{2} \frac{\partial}{\partial t} \overline{\phi'^2} + \overline{\mathbf{v}'\phi'} \cdot \nabla \bar{\phi} + \frac{1}{2} \bar{\mathbf{v}} \cdot \nabla \overline{\phi'^2} + \frac{1}{2} \nabla \cdot \overline{\mathbf{v}'\phi'^2} = \overline{D'\phi'}, \quad (13.67)$$

and where we may assume $\overline{D'\phi'} < 0$. If the mean flow is small and if the third-order term may be neglected then in a statistically steady state we have

$$\overline{\mathbf{v}'\phi'} \cdot \nabla \bar{\phi} \approx \overline{D'\phi'} < 0. \quad (13.68)$$

Therefore, on average, the flux of ϕ is downgradient in regions of dissipation, implying a positive average eddy diffusivity, and a balance is maintained between the downgradient flux of ϕ (which increases the variance) and dissipation. However, it should also be clear from (13.67) that if the turbulence is not statistically stationary, or if there is a mean flow, then downgradient transport cannot necessarily be expected. Indeed, the transport may be upgradient in regions where the eddy variance is falling, for then we may have the balance

$$\overline{\mathbf{v}'\phi'} \cdot \nabla \bar{\phi} \approx -\frac{1}{2} \frac{\partial}{\partial t} \overline{\phi'^2} > 0. \quad (13.69)$$

13.5 HOMOGENIZATION OF A SCALAR THAT IS ADVECTED AND DIFFUSED

Let us now assume that the effects of turbulence on a tracer are indeed diffusive. An important consequence of this is that, in the absence of additional forcing, there can be no extreme values of the tracer in the interior of the fluid and, in some circumstances, the diffusion will *homogenize* values of the tracer in broad regions. In this section we demonstrate and explore these properties.

13.5.1 Non-existence of extrema

Consider a tracer that obeys the equation

$$\frac{D\phi}{Dt} = \nabla \cdot (\kappa \nabla \phi) + S, \quad (13.70)$$

where $\kappa > 0$ and the advecting velocity is divergence-free. We now show that in regions where the source term, S , is zero there can be no interior extrema of ϕ if the flow is steady. The proof is in the form of a *reductio ad absurdum* argument — we first suppose there *is* an extrema of ϕ in the fluid, and prove a contradiction.

Given an extremum, there will then be a surrounding surface (in three dimensions), or a surrounding contour (in two), connecting constant values of ϕ . For definiteness consider two-dimensional incompressible flow for which the steady flow satisfies

$$\nabla \cdot (\mathbf{u}\phi) = \nabla \cdot (\kappa \nabla \phi). \quad (13.71)$$

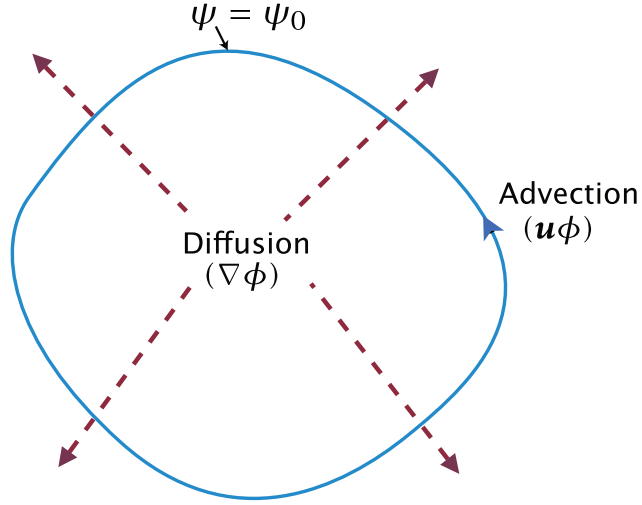


Fig. 13.5 If an extremum of a tracer ϕ exists in the fluid interior, then diffusion will provide a down-gradient tracer flux. But over an area bounded by a streamline, or by an isoline of ϕ , the net advective flux is zero. Thus, the diffusion cannot be balanced by advection and so in a steady state no extrema can exist.

Integrating the left-hand side over the area, A , enclosed by the iso-line of ϕ , and applying the divergence theorem, gives

$$\iint_A \nabla \cdot (\mathbf{u}\phi) dA = \oint (\mathbf{u}\phi) \cdot \mathbf{n} dl = \phi \oint \mathbf{u} \cdot \mathbf{n} dl = \phi \iint_A \nabla \cdot \mathbf{u} dA = 0. \quad (13.72)$$

where $\mathbf{n}$ is a unit vector normal to the contour. (If we were to integrate over an area bounded by a velocity contour, then $\mathbf{u} \cdot \mathbf{n} = 0$ and the integral would similarly vanish.) But the integral of the right-hand side of (13.71) over the same area is non-zero; that is

$$\iint_A \nabla \cdot (\kappa \nabla \phi) dA = \oint \kappa \nabla \phi \cdot \mathbf{n} dl \neq 0, \quad (13.73)$$

if the integral surrounds an extremum. This is a contradiction for steady flow. Hence, there can be no isolated extrema of a conserved quantity in the interior of a fluid, if there is any diffusion at all. The result (which applies in two or three dimensions) is kinematic, in that ϕ can be any tracer at all, active or passive. The physical essence of the result is the integrated effects of diffusion are non-zero surrounding an extremum, and cannot be balanced by advection. Thus, if the initial conditions contain an extremum, diffusion will smooth away the extremum until it no longer exists. This process, and the homogenization discussed below, are illustrated in Fig. 13.5.

13.5.2 Homogenization in two-dimensional flow

For two-dimensional flow we can obtain a still stronger result if we allow ourselves to make more assumptions about the strength and nature of the diffusion. The steady distribution of a scalar quantity being advected by an incompressible flow is governed by

$$J(\psi, \phi) = \nabla \cdot (\kappa \nabla \phi) + S, \quad (13.74)$$

where the terms on the right-hand side represent diffusion and source terms. Suppose that these terms are small, in the sense that the individual terms on the left-hand side nearly balance each other, so that

$$|J(\psi, \phi)| \ll \frac{U\phi}{L}. \quad (13.75)$$

This means we are in the high Peclet number limit ($P = UL/\kappa \gg 1$), and the dominance of advection suggests that any steady solution to (13.74) is of the form

$$\phi = G(\psi) + \mathcal{O}(P^{-1}), \quad (13.76)$$

where G is (for the moment) any function of its argument. Thus, isolines of ϕ are nearly coincident with streamlines, and

$$\nabla\phi \approx \nabla\psi \frac{d\phi}{d\psi}. \quad (13.77)$$

On integrating (13.74) over the area, A , bounded by some closed streamline, $\psi = \psi_0$ say, the left-hand side vanishes and we obtain

$$0 = \iint_A S \, dA + \oint_{\psi_0} \kappa \nabla\phi \cdot \mathbf{n} \, dl. \quad (13.78)$$

Using (13.77) then gives

$$\iint_A S \, dA = - \oint_{\psi_0} \kappa \frac{d\phi}{d\psi} \nabla\psi \cdot \mathbf{n} \, dl. \quad (13.79)$$

Since $d\phi/d\psi$ is constant along streamlines, and using $\mathbf{u} = \nabla^\perp \psi$, we have

$$\frac{d\phi}{d\psi} = - \frac{\iint_A S \, dA}{\oint_{\psi_0} \kappa \mathbf{u} \cdot d\mathbf{l}}. \quad (13.80)$$

This relationship determines ϕ as a function of ψ — that is, it determines $G(\psi)$ — in terms of the forcing and dissipation acting on the fluid. If the fluid is both unforced and inviscid, then a steady solution pertains when ϕ is an *arbitrary* function of ψ . If the source term S is zero, but dissipation is non-zero then the denominator of (13.80) is non-zero and therefore ϕ must be uniform: ϕ has been *homogenized*. The homogenization result also follows if we choose to integrate over an area surrounded by an isoline of ϕ , ϕ_0 say, but we leave this as a problem for the reader.

Interpretation

The homogenization result applies to a statistically steady flow in which the eddy transport of ϕ -stuff by the eddying motion may be parameterized diffusively, and in which there is an approximate functional relationship between mean ϕ and mean ψ . The first of these assumptions we have discussed at length in previous sections. The second requires that the diffusion must not be too strong, so that locally the tracer is conserved on fluid parcels. In the steady state the tracer is then a function of the streamfunction, the same function everywhere within the closed region.

Given these assumptions, the dynamics giving rise to homogenization is transparent: integrating round a contour of ψ or ϕ the effect of the advective terms vanish; the source (S) and the diffusion must balance each other, and if there is no source term there can be no tracer gradient. Put another way, the flow will circulate endlessly and steadily around the contours of ψ , which nearly coincide with contours of ϕ . Advection cannot alter the mean value of ϕ , so diffusion smooths out gradients within the closed contours, effectively *expelling* gradients of ϕ to the boundaries and forming a plateau of ϕ -values. Because extrema of ϕ are forbidden, the value of ϕ on the plateau cannot be a maximum or minimum: at the edge of the plateau the values of ϕ must fall somewhere, and rise somewhere else. The plateau can be a flat region etched out of a hillside, but a plateau on top of a butte is forbidden, for in that case diffusion would erode the butte down to the level of the surrounding land. Our derivation makes no distinction between a passive scalar like a dye and an active scalar, like potential vorticity. In reality, in the latter case the dynamics will further constrain the flow because the scalar distribution must be consistent with the velocity field that advects it, and this is particularly important in the dynamics of ocean gyres.